Hyeongju Lee

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Data & Backend Engineer — 29 months building production data pipelines and backend systems, plus 3D computer-vision research. Master's completion Mar. 2027; available from Apr. 2027. Military service completed.

Education

Kyushu Institute of Technology — Fukuoka, Japan
Master's, Computer Science & Systems Eng. (Creative Informatics), GPA 3.63/4.0 — *Apr. 2025 – Mar. 2027 (exp.)*

Hankyong National University — Anseong, South Korea
B.S. in Electrical, Electronic & Control Engineering, GPA 4.00/4.50 — *Mar. 2018 – Feb. 2025*

Industry Experience

Vision Inside — Seoul, South Korea
Data & Backend Engineer (Industrial Technical Personnel, Jul. 2022 – Jul. 2024) — *Jul. 2022 – Nov. 2024 (29 months)*

- **KISTI-commissioned public R&D value-creation platform (later SMART K2C), Aug. 2022 – Jun. 2024 — sole data engineer.** Built the collection layer: Selenium crawlers for Amazon and Danawa plus public-data-portal APIs for the Public Procurement Service, run on a 10–20-machine crawling farm feeding a central MySQL/MariaDB. Covered 10,985 Amazon categories (2.2M ASINs) and 2,481 Danawa categories, growing to **80M+ rows** (incl. 16M reviews and 42M procurement attribute records); later moved Amazon collection to the Rainforest API. Migrated the partner-built prototype (models 1–2, Java/Tomcat web app) to company servers and set up its Docker runtime.
- Productionized the platform's analysis models as FastAPI services with Jinja2/Tabulator admin dashboards: price-band K-means clustering (Dunn-index k) for competitor benchmarking, a daily KeyBERT/Sentence-Transformers batch mining customer-required quality keywords from reviews, and SQL matching of national R&D projects to companies (**248K links**). Wrote the audit deliverables (operator manual, data-build report), presented results in technical meetings with KISTI researchers, and **contributed to securing the KRW 300M main-phase contract**. Details: zoocasso.github.io/en/work/kisti
- **Genomic variant annotation & risk-report system (Kyung Hee Univ. College of Korean Medicine × a Canadian bio lab), Aug. 2022 – Sep. 2023 — sole developer.** Built a FASTQ → BWA → samtools → Picard → GATK variant-calling pipeline and a 4-stage SQL annotation join (dbSNP, GWAS Catalog, 1000 Genomes, UniProt) over **∼1 TB of reference data**, loaded via chromosome-wise splitting and chunked pandas/SQLAlchemy ingestion. Served rule-based GWAS risk scores, 26-population allele-frequency comparisons, and drug/herbal-medicine lookups through a FastAPI JSON API; deployed it with Docker through a Harbor registry I set up on a Korea IDC Ubuntu server (DNS/HTTPS), internal access only. Served as the **technical bridge between the Korean and Canadian teams**, working with the lab's designer.
  Details: zoocasso.github.io/en/work/genomics
- **Life-insurance data tooling (team of three), Aug. – Oct. 2023.** Loaded public life-insurance disclosure workbooks (16 product categories) into MySQL and served them through a FastAPI/Tabulator web viewer; packaged xlwings-based VBA execution and a PyQt desktop prototype; extracted a VBA pricing model's named ranges into MySQL.

Research Experience

Kyushu Institute of Technology — Fukuoka, Japan
Graduate Researcher, 3D imaging (advisor: Prof. Min-Chul Lee) — *Apr. 2025 – Present*

- **Four first-author papers** (ICMV 2026, ICCAS 2026, ITC-CSCC 2026 & 2025) on training-free light-field / integral-imaging reconstruction. Ray-based inverse mapping (RIM) redistributes angular to spatial samples with zero quantization error under orthographic conditions: **+2.39 dB PSNR** and +0.14 SSIM on 7 Stanford Lytro scenes, and **+0.11 mAP50** for a downstream YOLO detector. Details: zoocasso.github.io/en/research/rim
- Replaced iterative origin-point search with direct camera-parameter computation: **25.45 s → 0.31 s** on CPU and 0.0057 s on GPU (CuPy), with SSIM identical to the iterative method.
  Details: zoocasso.github.io/en/research/origin-point
- **Rotary Yoneyama Memorial Foundation Scholar** (2026), selected through a Japanese-language interview.

Commissioned Research with Hitachi — Fukuoka, Japan / Netherlands
Research Support Staff (part-time), image-data pipeline — *Apr. 2025 – Present*

- Rewrote an unmaintainable inherited codebase into an end-to-end Python pipeline — machine-vision camera control and capture, preprocessing into model-ready datasets — with a PyQt GUI that non-developer researchers run themselves.
- **Led two junior researchers** and **deployed the system on site in the Netherlands**; work conducted in Japanese and English.

Hankyong National University, 3D Image Processing Systems Lab Anseong, South Korea
Undergraduate Researcher (advisor: Prof. Myungjin Cho) *Aug. 2024 – Mar. 2025*

- First-author poster on origin-point optimization for volumetric computational reconstruction at Photonics Conference 2024 (Optical Society of Korea) — the precursor of the ITC-CSCC 2025 paper.

Selected Projects

3DOS: Real-time Vision Enhancement on an XR Headset | *C++, OpenCV, Unity* Nov. 2024

- **Led a five-person team** at an XR device hackathon: Fourier peplography for fog/smoke removal and photon counting for low-light recovery, written in C++/OpenCV as an Android native plugin (arm64) loaded by a Unity (IL2CPP) app on the Metalens 2 headset, using only the device camera.

WJMax Game Server (side project, team of two) | *FastAPI, Docker, AWS* Aug. 2023 – Feb. 2024

- Sole server developer for a community rhythm game: FastAPI on Docker (EC2) with RDS MySQL behind a Cloudflare proxy/WAF, and a play-log ingestion API serving ∼300 average and ∼2,000 peak concurrent players during updates. Details: zoocasso.github.io/en/work/wjmax

Smart Mirror (capstone, two-person team) | *Python, OpenCV, Raspberry Pi* 2022

- OpenCV LBPH face recognition and Kakao speech-to-text voice control serving per-user dashboards on a half-mirror display.

Depth Estimation via Correlation Filter & VCR (course project) | *MATLAB* 2022

- FFT-based correlation filters combined with volumetric computational reconstruction to recover depth from elemental images — the starting point of the master's research.

CNN in MATLAB (self-study) | *MATLAB* 2022

- Implemented an MNIST CNN (convolution, ReLU, max pooling, softmax) with backpropagation and momentum SGD, following the examples in Phil Kim's *MATLAB Deep Learning*.

Publications (first author)

[1] **Hyeongju Lee**, Myungjin Cho, Min-Chul Lee, "Angular-to-Spatial Information Redistribution via Ray-based Inverse Mapping," *18th Int. Conf. on Machine Vision (ICMV 2026)*, Budapest. (Accepted)

[2] **Hyeongju Lee**, Myungjin Cho, Min-Chul Lee, "Virtual Lens Array Synthesis from Multi-View Images via iRIM," *Int. Conf. on Control, Automation and Systems (ICCAS 2026)*, Sapporo. (Accepted)

[3] **Hyeongju Lee**, Myungjin Cho, Min-Chul Lee, "Accelerated Origin Point Optimization for Volumetric Computational Reconstruction," *ITC-CSCC 2026*, Bangkok. (Oral) Code

[4] **Hyeongju Lee**, Gilsu Yeo, Myungjin Cho, Min-Chul Lee, "Image Enhancement of Volumetric Computational Reconstruction Using Origin Point Optimization," *ITC-CSCC 2025*, Seoul. (Poster) Code

[D1] **Hyeongju Lee** et al., "Image Enhancement of Volumetric Computational Reconstruction Using Origin Point Optimization" (in Korean), *Photonics Conference 2024*, Optical Society of Korea. (Poster)

Skills & Additional

Languages: Python (primary), SQL, C++; Java, JavaScript, C#, MATLAB (basic–intermediate)
Data & Backend: FastAPI, SQLAlchemy, pandas, MySQL/MariaDB, Selenium, BeautifulSoup, Jinja2/Tabulator, openpyxl/xlwings, PyQt
ML & Vision: PyTorch, scikit-learn, OpenCV, NumPy, CuPy, KeyBERT/Sentence-Transformers, YOLO-based evaluation, Blender
Bioinformatics: BWA, samtools, Picard, GATK, PyVCF; dbSNP, GWAS Catalog, 1000 Genomes, UniProt
Infrastructure: Docker, Harbor, Linux (Ubuntu/WSL), DNS/HTTPS (Let's Encrypt), AWS (EC2, RDS), Cloudflare, Git
Certification: SQLD — SQL Developer, Korea Data Agency (Oct. 2023)
Spoken: Korean (native), Japanese (business conversational), English (working proficiency)
Military: Completed as Industrial Technical Personnel at Vision Inside (Jul. 2022 – Jul. 2024, 23 months)